

Written Homework 1

Problem Solving, Math 1000 – Summer 2016

Due Friday, May 27th

Problem 1 *For each of the following items below, draw an example of a graph that satisfies the given properties:*

- (a) A graph with exactly 8 vertices and exactly 3 bridges.*
- (b) A graph with exactly 7 vertices and no bridges.*
- (c) A graph with at least 10 vertices, at least 9 edges, and exactly 3 components.*

Problem 2 *Draw an example of a connected graph with exactly 9 vertices that is not simple but is regular and has two vertices with five degrees of separation.*

Problem 3 *Draw an example of a graph with up to 4 vertices that is not simple but is regular.*

Problem 4 *Draw a graph that has exactly 10 vertices, and has an Euler Path.*

Problem 5 *Draw a graph that has exactly 8 vertices, and has an Euler Circuit.*

Problem 6 *State Euler's sum of degrees theorem and briefly explain why this theorem is true (you may be casual with your phrasing in the explanation – I just want to see that you understand the concept on an intuitive level).*

Problem 7 *Draw any graph which has a Hamilton Circuit but not a Hamilton Path. However, if this is impossible, clearly and succinctly explain why.*

Problem 8 *Draw a connected, simple graph with six vertices which has no Euler Circuit, no Euler Path, no Hamilton Circuit, and no Hamilton Path. However, if this is impossible, clearly and succinctly explain why.*

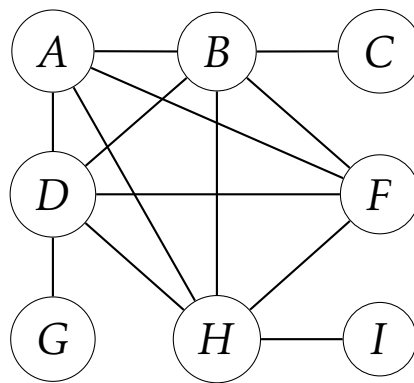
Problem 1

For each of the following items below, draw an example of a graph that satisfies the given properties:

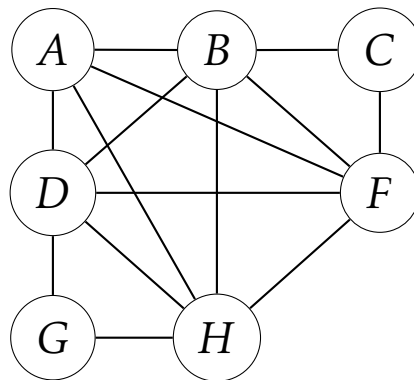
- (a) A graph with exactly 8 vertices and exactly 3 bridges.
- (b) A graph with exactly 7 vertices and no bridges.
- (c) A graph with at least 10 vertices, at least 9 edges, and exactly 3 components.

Solution

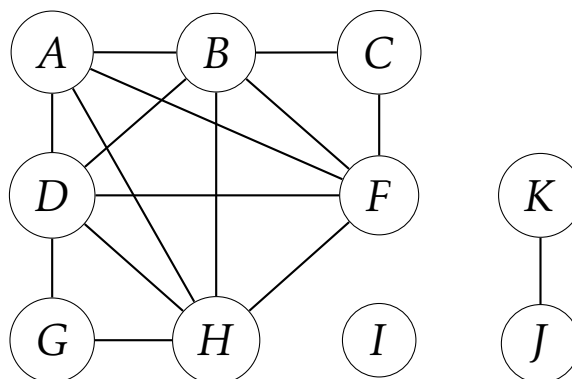
- (a) One example:



- (b) One example:



- (c) One example:

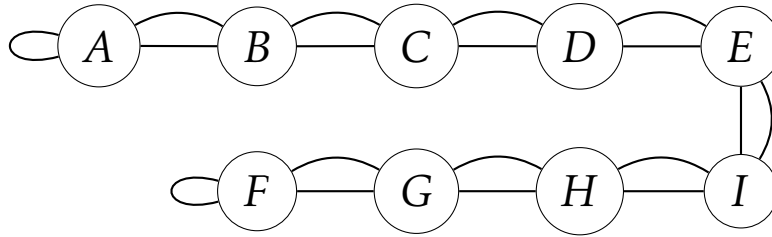


Problem 2

Draw an example of a connected graph with exactly 9 vertices that is not simple but is regular and has two vertices with five degrees of separation.

Solution

One example:

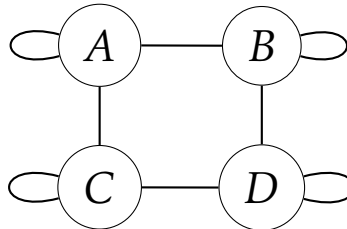


Problem 3

Draw an example of a graph with up to 4 vertices that is not simple but is regular.

Solution

One example:

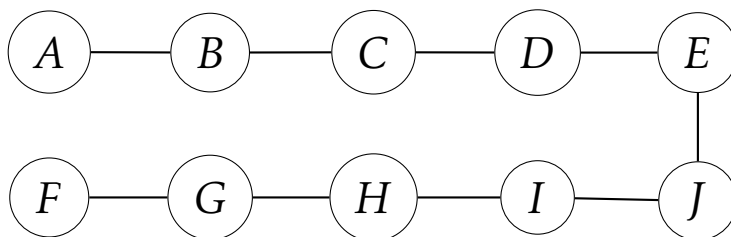


Problem 4

Draw a graph that has exactly 10 vertices, and has an Euler Path.

Solution

One example:

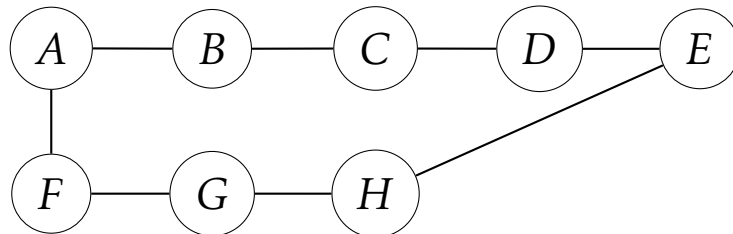


Problem 5

Draw a graph that has exactly 8 vertices, and has an Euler Circuit.

Solution

One example:

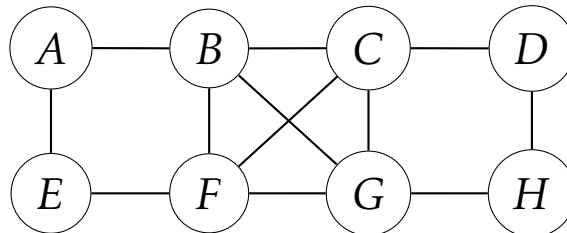


Problem 6

Draw a graph that has exactly 8 vertices, and has an Euler Circuit.

Solution

One example:



Problem 7

State Euler's sum of degrees theorem and briefly explain why this theorem is true (you may be casual with your phrasing in the explanation – I just want to see that you understand the concept on an intuitive level).

Solution

Euler's sum of degrees theorem states that the sum of the degrees of all the vertices in a graph is equal to twice the number of edges in the graph. This follows from the fact that each edge contributes two degrees – that is, it joins two vertices and increases each vertex's degree by one (or in a loop, joins a vertex to itself and increases its degree by two).

Problem 8

Draw any graph which has a Hamilton Circuit but not a Hamilton Path. However, if this is impossible, clearly and succinctly explain why.

Solution

This is impossible – any graph with a Hamilton circuit automatically has a Hamilton path. This follows from the fact that we may construct a Hamilton path in a graph with a Hamilton circuit by removing a single edge from the circuit (it doesn't matter which one, since it is a circuit).

Problem 9

Draw a connected, simple graph with six vertices which has no Euler Circuit, no Euler Path, no Hamilton Circuit, and no Hamilton Path. However, if this is impossible, clearly and succinctly explain why.

Solution

One example:

